

Introduction to Commutative Algebra

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CHAPTER 1

Lecture Notes

1. Lecture 1 (23/7)

CONVENTION. Rings are commutative with 1.

DEFINITION 2. Let R, S be rings. A *morphism* φ from R to S is a map $\varphi : R \rightarrow S$ such that

- (1) $\varphi(r + s) = \varphi(r) + \varphi(s)$,
- (2) $\varphi(rs) = \varphi(r)\varphi(s)$,
- (3) $\varphi(1) = 1$.

CONVENTION. A subring R must contain 1.

QUESTION. What is $R[X]$?

X is not an element of R . Consider

$$S = \{f : \mathbb{N} \rightarrow R \mid f(r) = 0 \text{ for all but finitely many } r\},$$

with the usual addition and multiplication in sequences (Cauchy product). Then S is a ring with identity

$$1_S = (1, 0, 0, \dots, 0).$$

Define $\varphi : R \rightarrow S$ by $\varphi(a) = (a, 0, \dots, 0)$. Then φ is a ring homomorphism, for sure.

Define $X = (0, 1, 0, \dots)$ and observe $X^2 = (0, 0, 1, 0, \dots)$. A polynomial P over R can then be identified as an element of S , say

$$P = (P_0, P_1, \dots, P_n, 0, \dots) = \varphi(P_0) \cdot 1_S + \varphi(P_1) \cdot X + \dots + \varphi(P_n) \cdot X^n.$$

DEFINITION 3. Let R be a ring. A non-zero, non-unit element $p \in R$ is *prime* if

$$p \mid ab \implies p \mid a \text{ or } p \mid b$$

(include the elementwise definition: for $a, b \in R$, $a \mid b$ if $\exists r \in R$ such that $b = r \cdot a$).

DEFINITION 4. Let R be a ring. A proper ideal P of R is called a *prime ideal* if

$$ab \in P \implies a \in P \text{ or } b \in P.$$

DEFINITION 5. Let R be a ring. $I \subseteq R$ is called an *ideal* if there exists a ring S and a morphism $\varphi : R \rightarrow S$ such that $I = \ker(\varphi)$.

REMARK 6. This immediately gives the elementwise definition of ideals.

$$\mathcal{N}(R) = \{\text{all nilpotent elements of } R\} \rightsquigarrow \text{the nil-radical of } R.$$

THEOREM 7.

$$\mathcal{N}(R) = \bigcap_{P \in \text{Spec } R} P, \quad \text{where } \text{Spec } R = \text{all prime ideals of } R.$$

PROOF. Let $x \in \mathcal{N}(R)$. Then for any $P \in \text{Spec } R$, there exists $n_P \in \mathbb{N}$ such that

$$x^{n_P} = 0 \implies 0 \in P \implies x^{n_P} \in P \implies x \in P.$$

Now let $x \notin \mathcal{N}(R)$, and define

$$S = \{I \subseteq R \text{ ideal} \mid x^n \notin I \text{ for all } n\}.$$

$S \neq \emptyset$, as $\langle 0 \rangle \in S$, and S is a poset (ordered by inclusion). By Zorn's Lemma, let M be a maximal element of S .

Claim. M is a prime ideal.

Proof of claim. Let $a \notin M$, $b \notin M$. Now,

$$M + \langle a \rangle, M + \langle b \rangle \notin S \implies \exists m, n \in \mathbb{N} \text{ such that}$$

$$x^m \in M + \langle a \rangle \text{ and } x^n \in M + \langle b \rangle \implies x^{m+n} \in M + \langle ab \rangle.$$

If $ab \in M$, then $x^{m+n} \in M$, a contradiction. Hence $ab \notin M$, so M is prime. □

Since $x \notin M$ and $M \in \text{Spec } R$, we conclude $x \notin \bigcap_{P \in \text{Spec } R} P$, completing the proof. □

QUESTION. What are the units of $R[X]$?

If R is an integral domain, then they are precisely the units of R .

PROPOSITION 8. $f(X) = a_n X^n + \cdots + a_1 X + a_0$ is a unit if a_0 is a unit and the a_i 's are nilpotent.

PROOF. Let $P \in \text{Spec } R$. We have a natural morphism $\varphi : R[X] \rightarrow (R/P)[X]$ given by

$$\varphi(b_m X^m + \cdots + b_1 X + b_0) = \overline{b_m} X^m + \cdots + \overline{b_1} X + \overline{b_0}, \quad (\overline{b_i} = b_i \text{ mod } P).$$

Then $\varphi(f)$ is a unit in $(R/P)[X]$, and since R/P is an integral domain, this forces

$$\overline{a_i} \equiv \overline{0} \pmod{P} \text{ for all } i \in [n], \quad \text{and} \quad \overline{a_0} \text{ is a unit of } R/P.$$

That is, $a_0 \notin P$ and $a_1, \dots, a_n \in P$. As $P \in \text{Spec } R$ was arbitrary,

$$a_0 \notin P \text{ and } a_i \in P \quad \forall i \in [n], \quad \text{for all } P \in \text{Spec } R.$$

The other direction is trivial. □

9. Lecture 2

DEFINITION 10. Let R be a ring and let $(M, +)$ be an abelian group. A *scalar multiplication*

$$R \times M \longrightarrow M, \quad (r, m) \longmapsto rm,$$

makes M into an *R -module* if the following hold for all $r, s \in R$ and $m, m_1, m_2 \in M$:

- (i) $(r + s)(m) = rm + sm$,
- (ii) $r(m_1 + m_2) = rm_1 + rm_2$,
- (iii) $r(sm) = rsm = s(rm)$,
- (iv) $1 \cdot m = m$.

DEFINITION 11. If M, N are R -modules, a map $\varphi : M \rightarrow N$ is called an *R -module morphism* if

- (i) $\varphi(m_1 + m_2) = \varphi(m_1) + \varphi(m_2)$,
- (ii) $\varphi(rm) = r\varphi(m)$.

DEFINITION 12. Let R be a ring and S a nonempty set. A *free R -module on S* is a pair (F, f) , where F is an R -module and $f : S \rightarrow F$ is a set-theoretic map, such that: for any R -module M and any set-theoretic map $g : S \rightarrow M$, there exists a unique R -module morphism $h : F \rightarrow M$ such that

$$\begin{array}{ccc} S & \xrightarrow{g} & M \\ f \downarrow & \nearrow h & \\ F & & \end{array}$$

commutes, i.e. $h \circ f = g$.

PROPOSITION 13. Let (F, f) be a free R -module on S . Then:

- (1) f is injective;
- (2) $\text{Im}(f)$ generates F as an R -module.

PROOF OF (1). Let $x, y \in S$ with $x \neq y$. **Goal:** $f(x) \neq f(y)$.

Take any R -module M with at least two elements, and define $g : S \rightarrow M$ such that $g(x) \neq g(y)$. By definition, there exists a (unique) morphism $h : F \rightarrow M$ such that $h \circ f = g$. Then

$$(h \circ f)(x) = g(x) \neq g(y) = (h \circ f)(y) \implies f(x) \neq f(y). \quad \square$$

DEFINITION 14. Let M be an R -module and $\emptyset \neq A \subseteq M$. The *submodule of M generated by A* is, equivalently:

- (1) the smallest submodule of M containing A ;
- (2) the intersection of all submodules of M containing A ;
- (3) the set of all finite R -linear combinations of elements of A .

PROOF OF (2). Let A be the submodule of F generated by $\text{Im}(f)$. Since $f(s) \in \text{Im}(f) \subseteq A$ for every $s \in S$, f factors as $f = i_A \circ f^*$ for a set-theoretic map $f^* : S \rightarrow A$ and the inclusion $i_A : A \hookrightarrow F$:

$$\begin{array}{ccc} S & \xrightarrow{f^*} & A \\ & \searrow f & \downarrow i_A \\ & & F \end{array}$$

As (F, f) is free on S and A is an R -module, there is a unique morphism $h : F \rightarrow A$ such that $h \circ f = f^*$:

$$\begin{array}{ccc} S & \xrightarrow{f^*} & A \\ f \downarrow & \nearrow h & \\ F & & \end{array}$$

It is enough to show i_A is onto. We compute

$$(i_A \circ h) \circ f = i_A \circ (h \circ f) = i_A \circ f^* = f.$$

So $i_A \circ h$ and id_F both satisfy $(-) \circ f = f$; by uniqueness in the definition of freeness (applied with $M = F$, $g = f$), $i_A \circ h = \text{id}_F$. Hence i_A is onto, i.e. $A = F$, so $\text{Im}(f)$ generates F . $\square$

THEOREM 15 (Uniqueness of free modules). Let (F, f) be a free R -module on a set S . Then (F', f') is also a free R -module on S if and only if there exists a unique R -module isomorphism $j : F \rightarrow F'$ such that $j \circ f = f'$.

PROOF. Suppose (F', f') is also free on S . Since (F, f) is free (applied to $M = F'$, $g = f'$), there is a morphism $h : F \rightarrow F'$ with $h \circ f = f'$:

$$\begin{array}{ccc} S & \xrightarrow{f'} & F' \\ f \downarrow & \nearrow h & \\ F & & \end{array}$$

Likewise, since (F', f') is free (applied to $M = F$, $g = f$), there is a morphism $h' : F' \rightarrow F$ with $h' \circ f' = f$:

$$\begin{array}{ccc} S & \xrightarrow{f} & F \\ f' \downarrow & \nearrow h' & \\ F' & & \end{array}$$

Then $(h' \circ h) \circ f = h' \circ (h \circ f) = h' \circ f' = f$. So $h' \circ h$ and id_F both satisfy $(-) \circ f = f$; by uniqueness (with $M = F$, $g = f$), $h' \circ h = \text{id}_F$. Similarly $h \circ h' = \text{id}_{F'}$. So $j := h$ is an isomorphism with $j \circ f = f'$.

Converse. Suppose there is an isomorphism $j : F \rightarrow F'$ with $j \circ f = f'$. To show (F', f') is free on S , take any R -module M and set-theoretic map $g : S \rightarrow M$; we must produce a unique $h' : F' \rightarrow M$ with $h' \circ f' = g$:

$$\begin{array}{ccc} S & \xrightarrow{g} & M \\ f' \downarrow & \nearrow h' & \\ F' & & \end{array}$$

Since (F, f) is free, there is a unique $h : F \rightarrow M$ with $h \circ f = g$. From $j \circ f = f'$ we get $j^{-1} \circ f' = f$, so

$$(h \circ j^{-1}) \circ f' = h \circ (j^{-1} \circ f') = h \circ f = g,$$

i.e. $h' := h \circ j^{-1}$ works:

$$\begin{array}{ccc} S & \xrightarrow{g} & M \\ f' \downarrow & \searrow f & \uparrow h \\ F' & \xrightarrow{j^{-1}} & F \end{array}$$

Uniqueness. Suppose $t : F' \rightarrow M$ also satisfies $t \circ f' = g$. Then $t \circ j : F \rightarrow M$ satisfies $(t \circ j) \circ f = t \circ (j \circ f) = t \circ f' = g$, so by uniqueness of h (from freeness of (F, f)), $t \circ j = h$, hence $t = h \circ j^{-1} = h'$. $\square$

THEOREM 16 (Existence of free R -modules). *For any ring R and nonempty set S , a free R -module on S exists.*

PROOF. Define

$$F = \{\theta : S \rightarrow R \mid \theta(s) = 0 \text{ for almost all } s \in S\}.$$

For $\theta_1, \theta_2 \in F$ and $\lambda \in R$, define

$$(\theta_1 + \theta_2)(s) = \theta_1(s) + \theta_2(s), \quad (\lambda\theta)(s) = \lambda\theta(s).$$

One checks that F is an R -module. Define $f : S \rightarrow F$ by $f(s) = \delta_s$, where $\delta_s(t) = 1$ if $t = s$ and 0 otherwise.

Existence of h . Let M be an R -module and $g : S \rightarrow M$ a set-theoretic map. Define $h : F \rightarrow M$ by

$$h(\theta) = \sum_{s \in S} \theta(s) g(s),$$

a finite sum for every $\theta \in F$. For any $t \in S$,

$$(h \circ f)(t) = h(f(t)) = h(\delta_t) = \delta_t(t) g(t) = g(t),$$

so $h \circ f = g$:

$$\begin{array}{ccc} S & \xrightarrow{g} & M \\ f \downarrow & \nearrow h & \\ F & & \end{array}$$

Uniqueness of h . For $\theta \in F$ and $t \in S$,

$$\theta(t) = \theta(t) \cdot 1_R = \sum_{s \in S} \theta(s) \delta_s(t) = \left(\sum_{s \in S} \theta(s) \delta_s \right) (t) \quad \forall t \in S,$$

so $\theta = \sum_{s \in S} \theta(s) \delta_s$. Now let $h' : F \rightarrow M$ be any R -module morphism with $h' \circ f = g$. Then for $\theta \in F$,

$$h'(\theta) = h' \left(\sum_{s \in S} \theta(s) \delta_s \right) = \sum_{s \in S} \theta(s) h'(\delta_s) = \sum_{s \in S} \theta(s) h'(f(s)) = \sum_{s \in S} \theta(s) g(s) = h(\theta).$$

Hence $h = h'$. □

CHAPTER 2

Problem Discussions

Bibliography